Predictive Modeling

Objective: To extract actionable insights and performed to understand the drivers of sales and build a predictive model.

Data Description

- The dataset contains 738 firms.
- There are 9 variables
- Sales: sales (Normalized values).
- Infrastructure: capital (Net stock of property and equipment) and employment (Number of employees).
- Property: patents (Granted patents) and randd (R&D stock).
- Market Indicators: sp500 (Index membership), tobinq (Asset value ratio), value (Market value), and institutions (Institutional ownership).

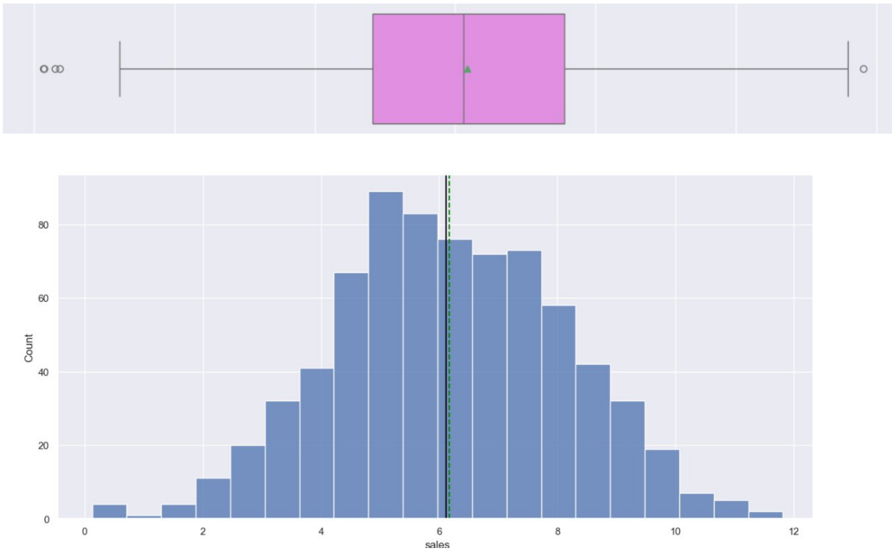
Observations	Variable	Duplicate
738	9	0

- All variables are float, exclude patents (int) and sp500 (object).

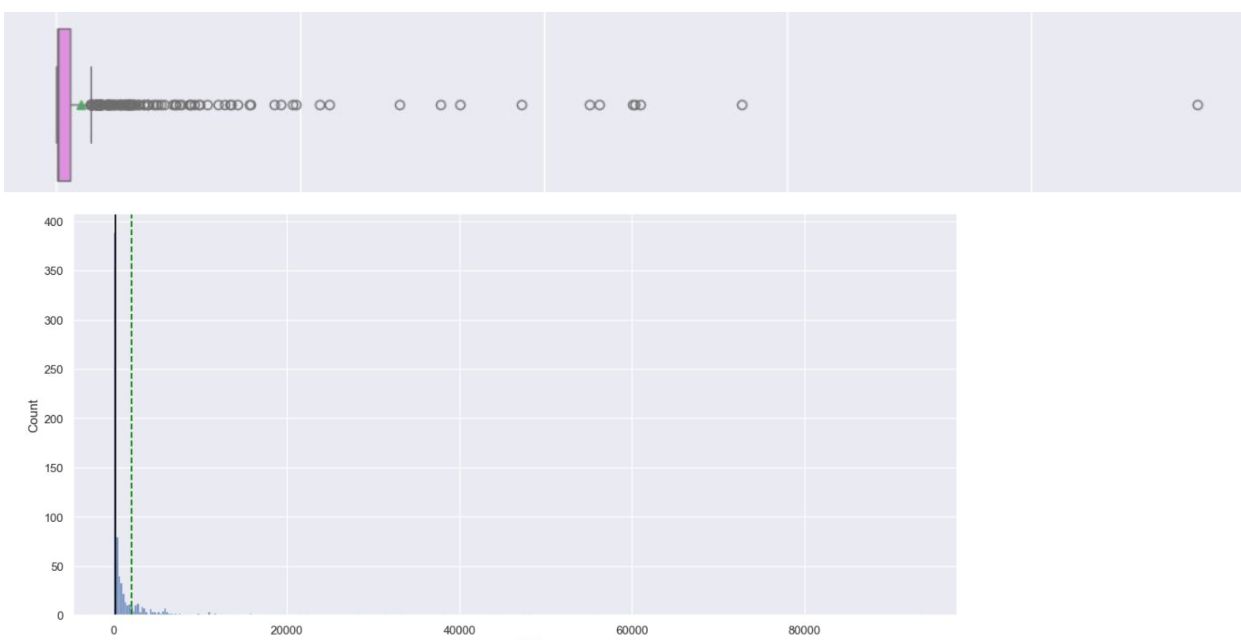
Content

Based on the univariate analysis using the combined histogram and boxplot visualization for each variable

- **Sales**:
- The distribution of sales is relatively symmetrical and appears to have been log-transformed. The mean (6.17) and median (6.11) are very close.
- There are a few outliers on both the lower and higher ends, but the data is largely concentrated between 4 and 8.
- range is small (0.13 to 11.8)



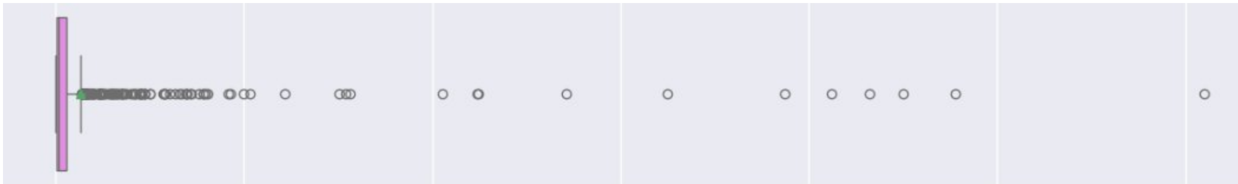
- **Capital**
- Extreme right skewness distrubution.
- The median is only 205.8, while the mean is 2,028.5, driven by massive outliers (max: 93,625).
- Small-to-medium scale, while a handful of "mega-firms" dominate the total capital and workforce metrics

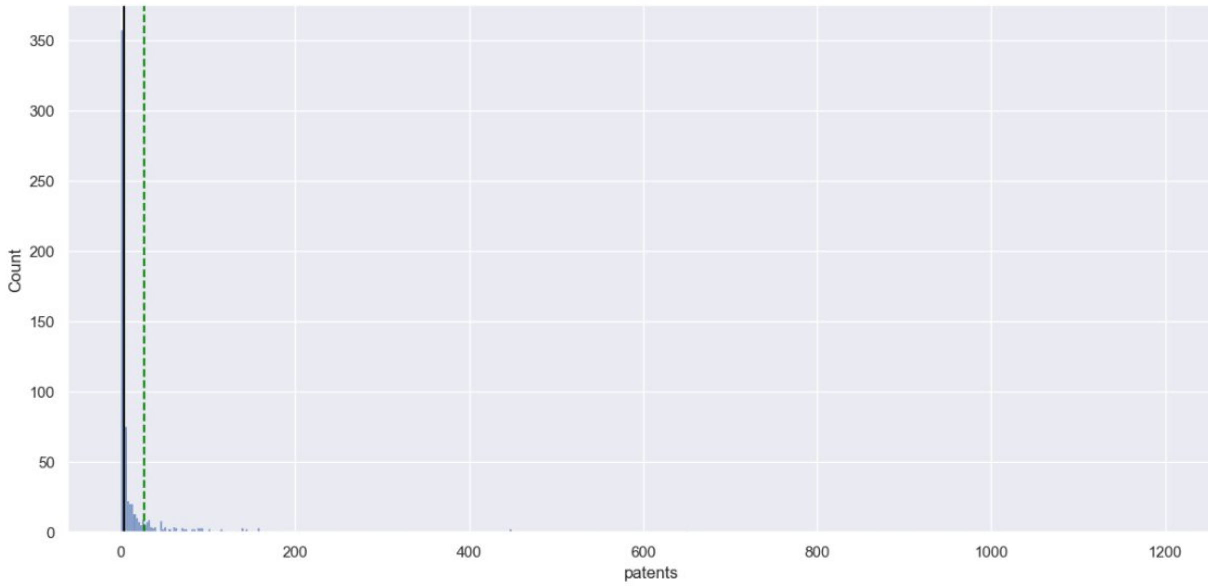


➢ **Patents:**

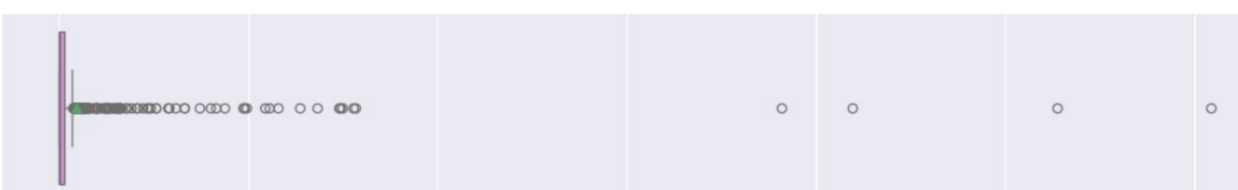
- Highly skewed to the right.

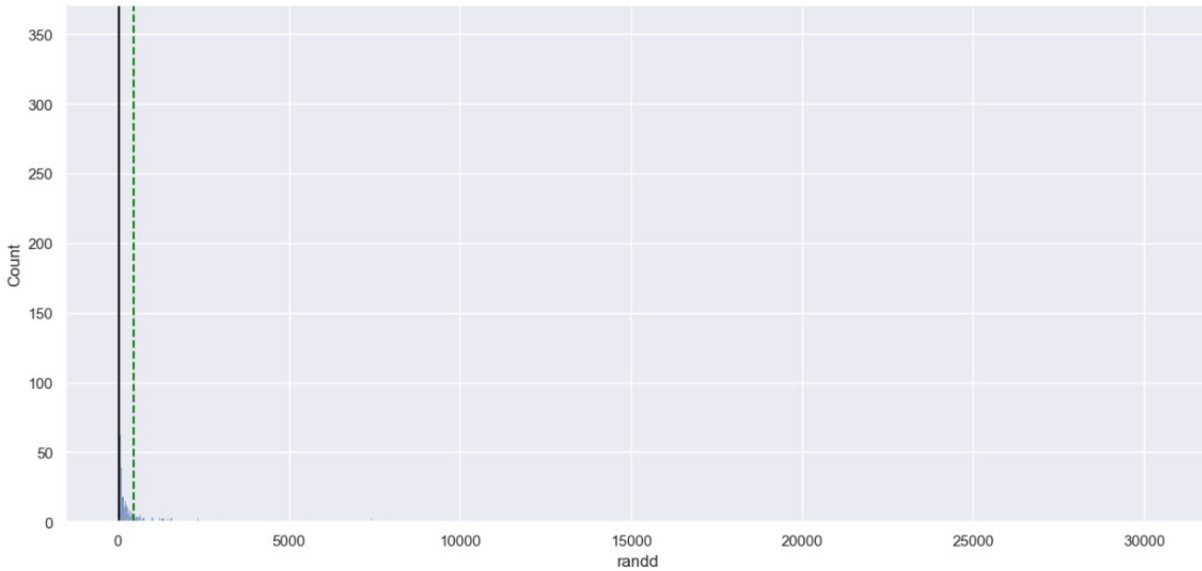
- The median number of patents is just 3, while the mean is 26.3. The maximum is 1,220.
- Innovation is highly concentrated.





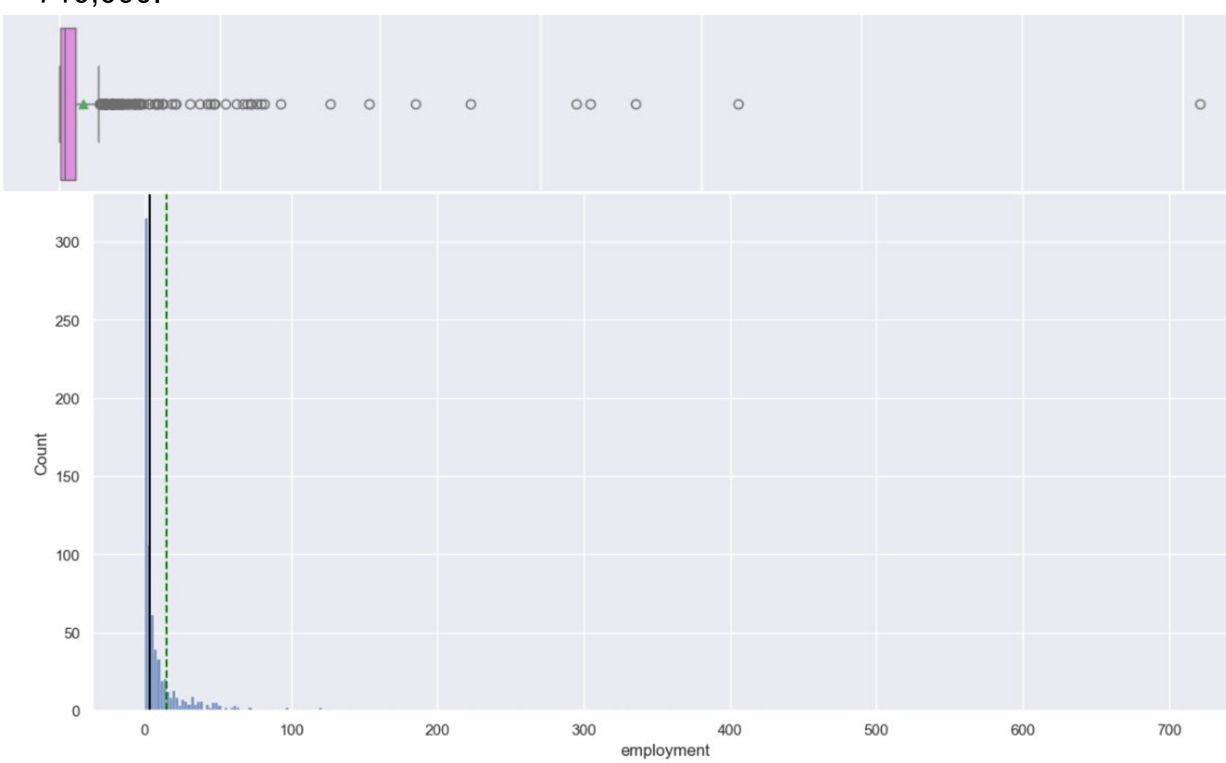
- Randd
  - Exhibit extreme right skewness.
  - Most firms spend relatively little (Median: 36.8), but a few "innovation giants" spend up to 30,425.





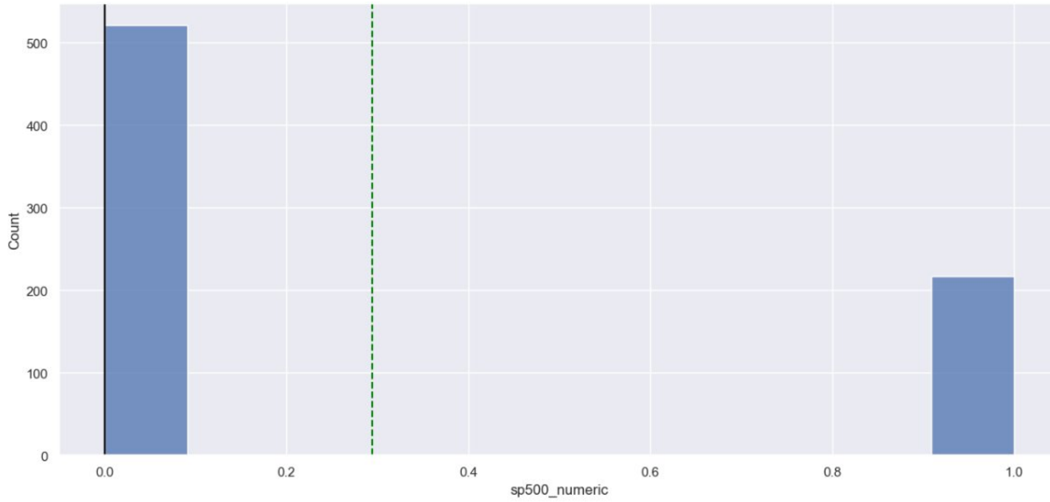
- Employment

  - right skewness.
  - The median firm has about 3,000 employees, but the largest employs over 710,000.



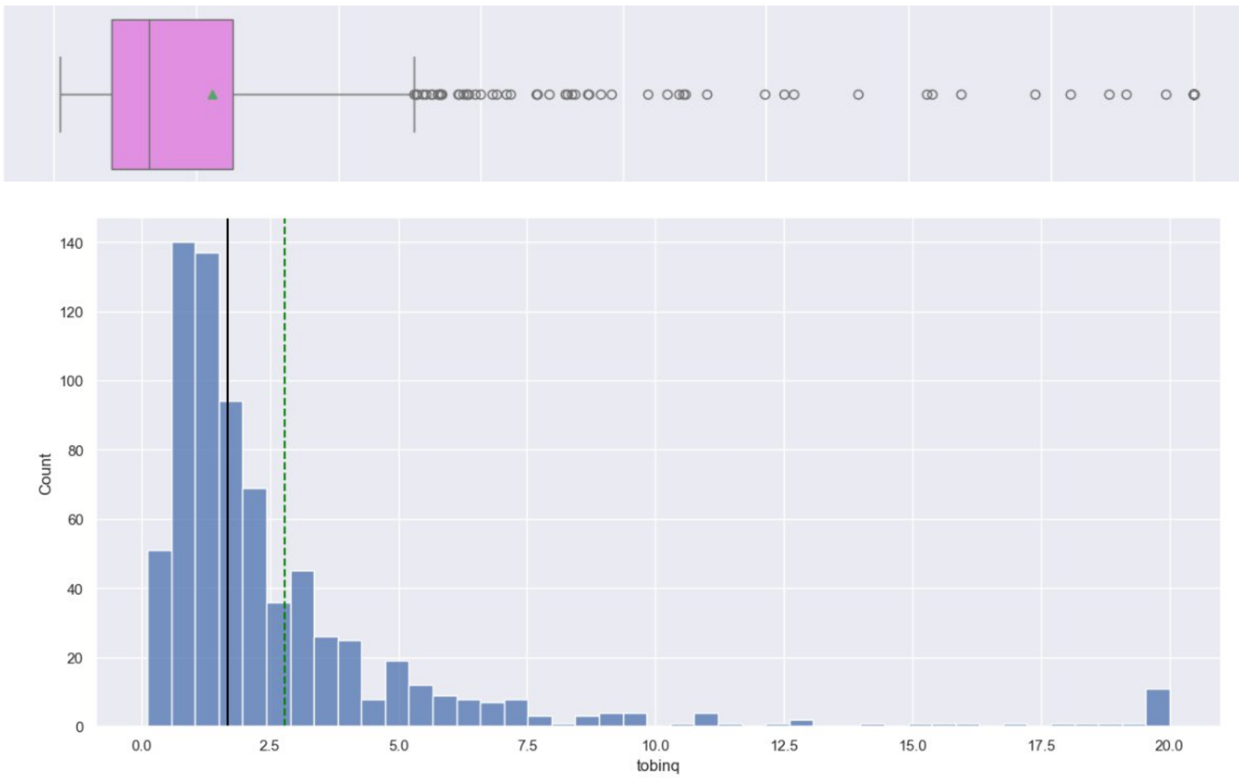
- **SP 500 Membership**
  - The majority of firms in the dataset are not in the SP 500.

- Heavy right-skewness in other variables;
- The non-SP 500 firms form of the distribution at the lower end, while the SP 500 firms constitute the long tails (outliers) in terms of value, capital.

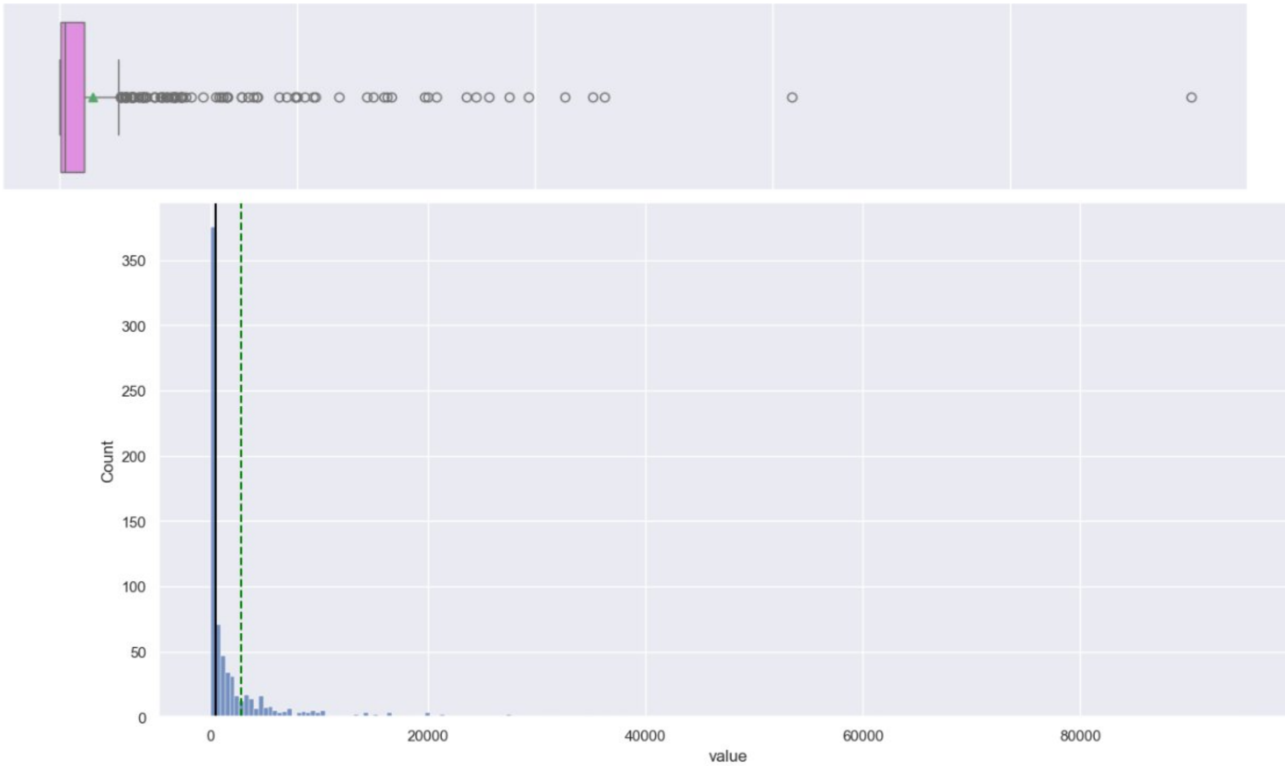


➢ **Tobinq**

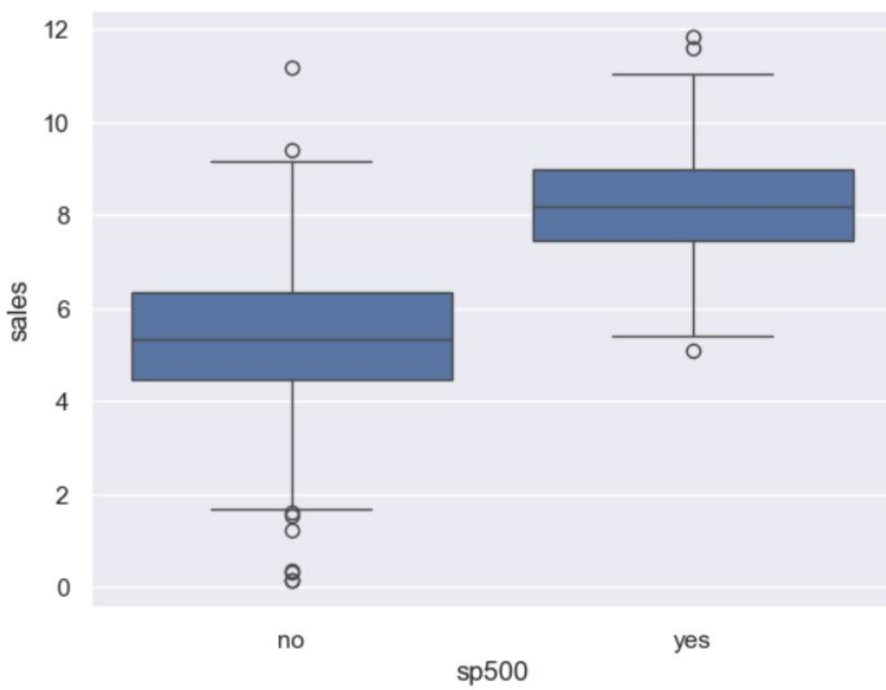
- Right-skewed with a heavy concentration between 1.0 and 3.0.
- The mean (2.79) is higher than the median (1.68). There are significant outliers on the higher side
- The high-end outliers represent firms that the market values extremely highly relative to their physical size.



- **Value**
- right skewed.
- The median market value is $418M, but the mean is $2.79B.
- Insight: The Market Valueis heavily by the top 10-25% of firms.

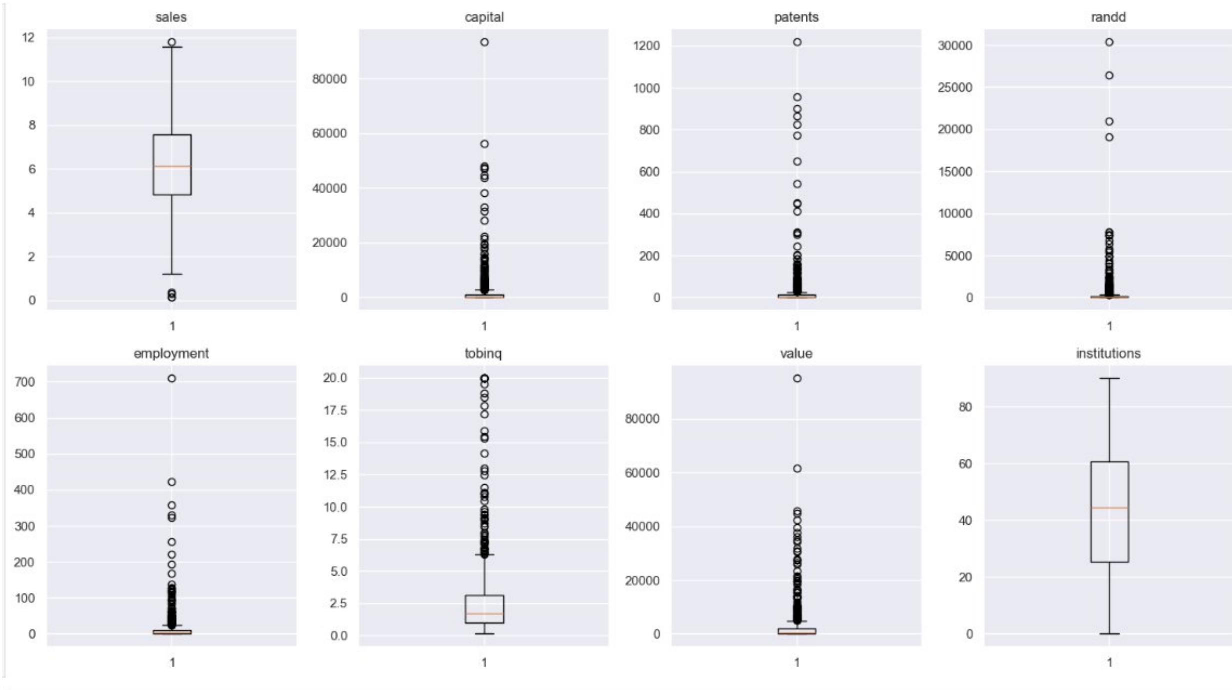


Comparison of sp500 vs sales:



- Distribution Comparison: You can see if companies in the S&P 500 (usually 1) have significantly higher median sales than those that are not (0).
- Outliers: The dots beyond the whiskers identify "Superstar" companies whose sales are far beyond the industry norm.
- Skewness: If the median line is closer to the bottom of the box, the sales data for that group is right-skewed (a few companies have very high sales).

- **Patents & randd (0.82):** This is your strongest relationship.
- **Capital & randd (0.77):** Significant capital investment is strongly tied to randd expenditure.
- **Employment & randd (0.78):** randd intensive firms are also high-employment firms. labor-intensive activity in this dataset.
- There is a strong correlation between **value and capital (0.72)**.Market value is closely tied to capital assets.



➢ Sales and Tobin's Q are-0.29: Larger companies (by sales) tend to have lower asset-premium ratios.
➢ zero correlation with value (-0.01) and institutions (-0.03

Preparation for the regression model:

- Independent variables (randd, capital) were zero.
- Extreme Skewness: patents and randd are highly right-skewed,
- High Variance: tobinq.
- very high correlation (0.82) between patents and randd.
- Employment and Capital: These variables also move closely with RandD spending (0.78 and 0.77 respectively).
- coefficient for sp500_yes will represent the average difference in sales for an SP 500 company compared to the baseline, holding all other variables constant.

```
       capital  patents        randd  employment sp500     tobinq  \
0   161.603986       10   382.078247    2.306000    no  11.049511
1   122.101012        2     0.000000    1.860000    no   0.844187
2  6221.144614      138  3296.700439   49.659005   yes   5.205257
3   266.899987        1    83.540161    3.071000    no   0.305221
4   140.124004        2    14.233637    1.947000    no   1.063300

          value  institutions
0   1625.453755         80.27
1    243.117082         59.02
2  25865.233800         47.70
3     63.024630         26.88
4     67.406408         49.46
0    6.719007
1    6.013113
2    9.037039
3    6.113682
4    5.170075
Name: sales, dtype: float64
```

	const	capital	patents	randd	employment	tobinq	value	institutions	sp500_yes
0	1.0	161.603986	10.0	382.078247	2.306000	11.049511	1625.453755	80.27	0.0
1	1.0	122.101012	2.0	0.000000	1.860000	0.844187	243.117082	59.02	0.0
2	1.0	6221.144614	138.0	3296.700439	49.659005	5.205257	25865.233800	47.70	1.0
3	1.0	266.899987	1.0	83.540161	3.071000	0.305221	63.024630	26.88	0.0
4	1.0	140.124004	2.0	14.233637	1.947000	1.063300	67.406408	49.46	0.0

Variable	**Type**
const	Numeric (1.0)
capital, randd, employment	Numeric (High Variance)
patents	Numeric (Highly skewed)
sp500_yes	Binary Dummy (0 or 1)

- **Data Splitting: split the data in 80:20 ratio**

Number of rows in train data = 590
Number of rows in test data = 148

- **Performance evaluation:**

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  sales   R-squared:                       0.669
Model:                            OLS   Adj. R-squared:                  0.664
Method:                 Least Squares   F-statistic:                     146.7
Date:                Sun, 11 Jan 2026   Prob (F-statistic):          4.33e-134
Time:                        21:19:04   Log-Likelihood:                -907.85
No. Observations:                 590   AIC:                             1834.
Df Residuals:                     581   BIC:                             1873.
Df Model:                           8
Covariance Type:            nonrobust
------------------------------------------------------------------------------
```

- R-squared of 0.669, meaning the independent variables explain approximately 66.9% of the variance
- Adjusted R-squared is 0.664, which is very close to the R-squared value.
- The Prob (F-statistic) is extremely low.

- **Checking model performance on train set (seen 70% data)**

Training Performance

	RMSE	MAE	MAPE
0	1.127269	0.844911	26.95745

- **MAE (0.845):** On average, model's predictions deviate from the actual sales value by **0.845 units**.
- **RMSE (1.127):** MSE is noticeably higher than the MAE that the model is being penalized by **large errors** on specific observations.
- **MAPE (26.96%):** This indicates that, on average, the model's predictions are off by roughly **27%.**

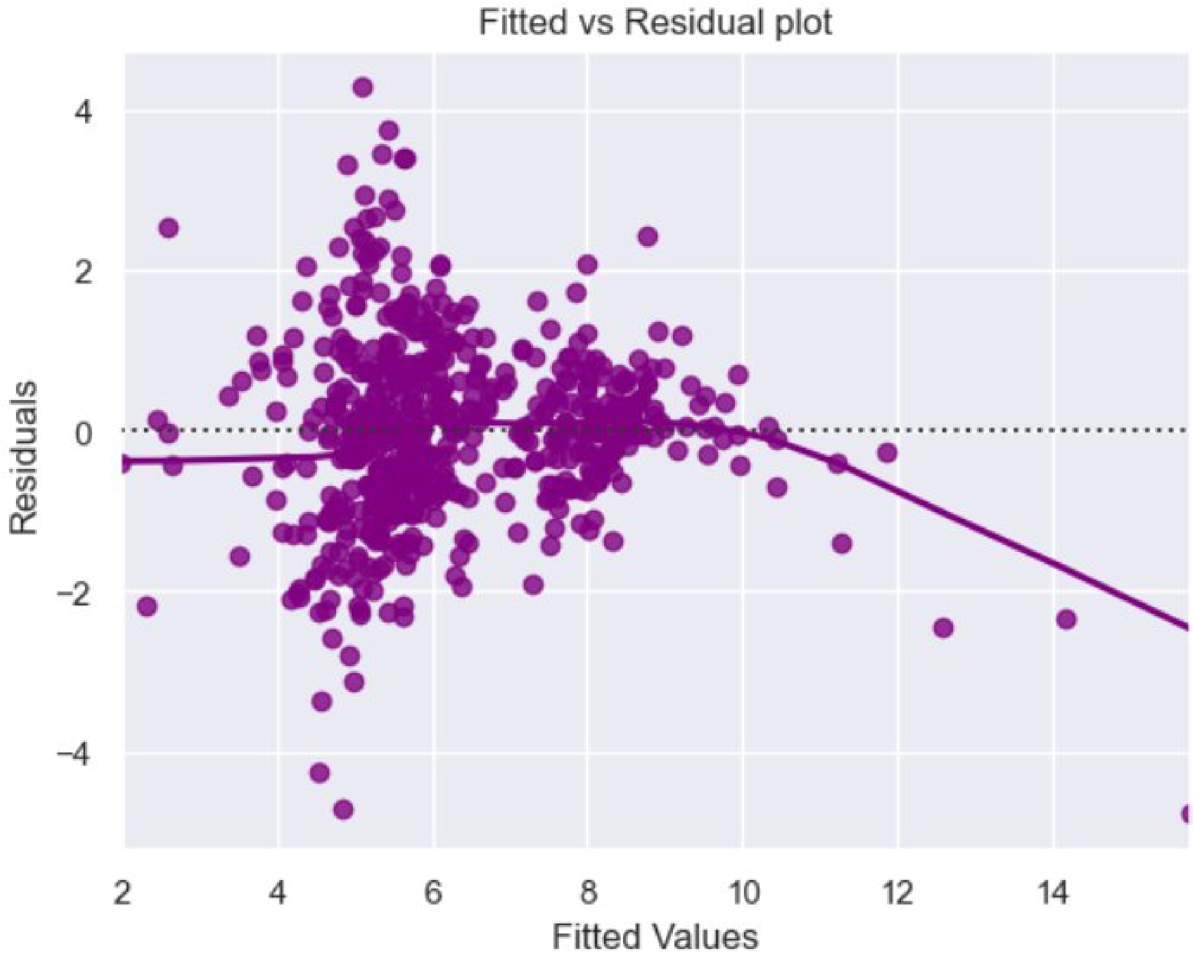
➢ **Checking model performance on test set (seen 30% data)**

Test Performance

	RMSE	MAE	MAPE
0	1.030785	0.81383	23.978281

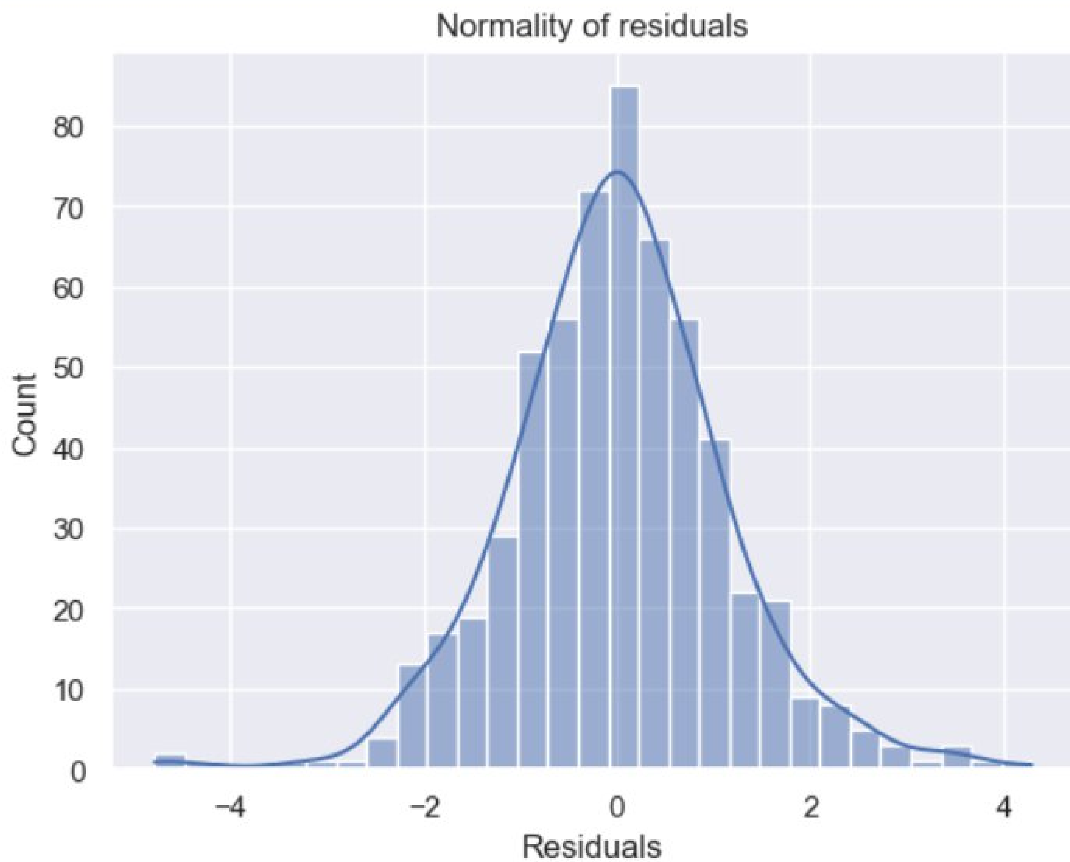
- **RMSE (1.031)**: The Root Mean Squared Error on the test set is lower than the training RMSE (1.127 from previous steps), showing the model handles unseen data well.
- **MAE (0.814)**: On average, test predictions are off by roughly **0.814 units**.
- **MAPE (23.98%)**: The Mean Absolute Percentage Error indicates that predictions are typically within **24%** of the actual sales values.

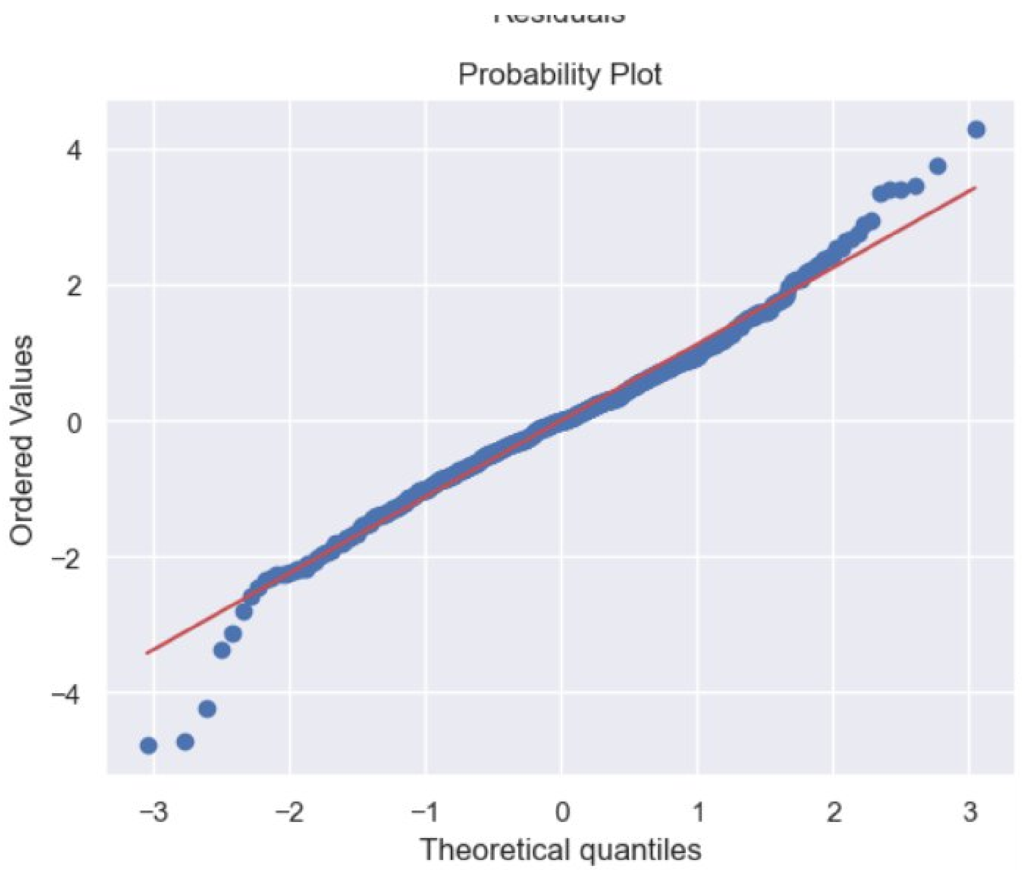
Fitted values vs residuals



Statistical Significance: The **F-statistic (146.7)** and its extremely low probability which confirm that the overall regression model is highly significant.

TEST FOR NORMALITY


Normality of residuals
80
70
60
50
40
30
20
10
0
Count
−4
−2
0
2
4
Residuals



- The **Histogram** shows a generally bell-shaped distribution centered near zero, which is a positive sign for normality.

- **Significant Deviation:** Observation **366** shows a much larger error with a residual of **0.944**, indicating an under-prediction of nearly 1 unit.

- The **Fitted vs. Residual plot** shows a downward-curving purple trend line at higher fitted values, suggesting the relationship might be non-linear.